RESEARCH REPORT / 11 NOVEMBER 2026

Pilot
A 63.57M-token Transformer comparison

Nonlinear matrix memory, FFT-compatible exponential history and an exploratory language-model evaluation

Two parameter-matched decoder language models were trained on the same 63,569,920 scored FineWeb-Edu next-token targets. Pilot uses a separate exponential input history and a nonlinear, two-gate matrix recurrence. Its FFT and scan implementations are tested as alternative evaluations of the same history, not as separately trained architectures.

63.57M scored training targets per model	**16.7%** lower test perplexity for Pilot
105.15 / 126.24 Pilot / Transformer test perplexity	**~60×** slower Pilot training throughput

Finding. Pilot achieves better held-out language prediction in this matched-token run, and its alternative history backends pass numerical checks. Neither model demonstrates reliable zero-shot factual retrieval. Pilot shows little useful long-distance copying and is substantially slower to train. The results justify further testing, not a general superiority or long-memory claim.

Rickey1990 research project • Pilot (formerly Matrix EXILRM) • Single-seed exploratory study. Report based on the supplied run archive; not an independent replication.

Abstract and research question

Pilot investigates whether explicit exponential history can guide an expressive recurrent matrix state without attention. The experiment compares the original fixed-decay design against a standard causal Transformer, retaining ordinary next-token training and matching total parameter count and training exposure. No task-specific supervision or external key/value hints are used.

Both models complete 15,520 optimiser updates. Pilot reaches 4.6554 test NLL versus 4.8382 for the Transformer, corresponding to a 16.7% reduction in perplexity. However, the recurrent implementation trains approximately 60 times more slowly, and neither model learns dependable retrieval on the small evaluation-only probe suite.

1. Model configurations

Configuration	Transformer	Pilot
Trainable parameters	428,608	428,523
Blocks / width	3 / 64	3 / 64
Heads / head dimension	4 / 16	8 / 8
FFN hidden width	256	289
Position / history	Learned positions	Exponential (λ = 0.9)
Persistent state	Attention over context	8 matrices of 8 × 8 per layer
Readout	Causal attention	Matrix × query

Both models use pre-normalised residual blocks, GELU feed-forward networks, a final layer norm and tied input/output token embeddings. Pilot has no positional embedding table, no local attention and no added time-mixing block. Its FFN is wider to match total parameter count; the difference is 85 weights (0.020%).

Interpretation of the design

Equal parameter count and scored targets do not imply equal compute. Pilot FFT is the fully trained configuration. Pilot Scan and Pilot Parallel are numerical implementations of the same state equations, evaluated through short calibration runs and a final-checkpoint backend audit. No independent 63.57M scan-trained endpoint is available.

2. Architecture and mathematics

For each layer, let z_t be the normalised token representation. Learned projections produce key k_t, value v_t and query q_t in each head. With head dimension d = 8, the input is a matrix:

$$X_t = v_t\, k_t^T$$

Explicit history: input-only, strictly causal

$$P_0 = 0; \quad P_t = [\Sigma_{j=1}^{t} \lambda^{j-1} X_{t-j}] \,/\, [\Sigma_{j=1}^{t} \lambda^{j-1}]$$

The history is computed from prior input matrices, never prior hidden matrices. For fixed λ, the numerator is a causal convolution with kernel [0, 1, λ, λ², ...]. Zero-padding to at least 2T−1, FFT multiplication, inverse FFT and cropping give the numerator. Dividing by the cumulative kernel weights yields P.

Nonlinear state correction

Let L denote an independently learned row/column transformation $L(A) = UA + AV^T$ (with a bias on the current-input path). Each gate has separate transformations of X, P and M. The following equations suppress head indices; M starts at zero.

$$R_t = \text{sigmoid}(L_{rX}(X_t) + L_{rP}(P_t) + L_{rM}(M_{t-1}))$$

$$W_t = \text{sigmoid}(L_{wX}(X_t) + L_{wP}(P_t) + L_{wM}(M_{t-1}))$$

$$C_t = \tanh(LcX(X_t) + LcP(P_t) + LcM(R_t \odot M_{t-1}))$$

$$M_t = M_{t-1} + W_t \odot (C_t - M_{t-1})$$

$$y_t = M_t\, q_t \,/\, \sqrt{d}$$

Head reads are concatenated, projected and added to the residual stream, followed by the FFN. The learned state is not constrained to be Toeplitz, diagonal or circulant. Matrix mixing costs $2d^2$ weights per map (128 at d = 8), excluding biases, rather than d^4 for a dense flattened-state map.

FFT, sequential scan and parallel prefix

$$S_{t+1} = X_t + \lambda S_t; \quad D_{t+1} = 1 + \lambda D_t; \quad P_t = S_t / D_t$$

The same history admits a sequential scan (zero initial S and D; $P_0 = 0$) or a parallel prefix via affine composition. FFT exploits implicit Toeplitz structure in time without materialising a T × T matrix. This standard convolution property applies to the history only: the state-dependent nonlinear gates remain sequential.

Notation: ⊙ denotes elementwise multiplication, not matrix multiplication. The row/column maps are independent across gates, input paths and layers; each map is shared across the eight heads within its layer. See the block diagram on page 10.

3. Dataset and experimental procedure

FineWeb-Edu (sample-10BT), local Parquet shard 000_00000.parquet. The recorded manifest groups exact duplicate documents by hash and uses document-separated train, validation and test splits. The shared SentencePiece BPE tokenizer is fit on a training-only sample of approximately 5M characters.

Setting	Recorded value
Scored training targets	63,569,920 per model; 127,139,840 total
Prepared token files	100,000,001 train; 32,768 validation; 32,768 test
Vocabulary / context / batch	4,096 pieces / 256 tokens / 16 sequences
Optimiser	AdamW; β = (0.9, 0.95); weight decay 0.01; clip 1.0
Learning-rate schedule	0.00075 peak; 3,178,496-target warmup; cosine to 0.00005
Random seed	20261109 (one seed)
Training precision	FP32 for both; float64 history reference checks
Hardware / scheduling	Two Tesla T4 GPUs; one model per GPU
Final LM evaluation	8,192 scored positions each, validation and test
Checkpoint interval	3,178,496 scored targets; 20 intervals

The time-budget runner measures short disposable training benchmarks, verifies each backend against eager execution, then selects a common target and restarts both models from seeded initialisation. Training uses the same contiguous token windows with 4,096 scored targets per update. Recurrent states reset for each 256-token sequence; the Transformer uses a learned 256-position table.

Reproducibility boundary

The archive records completion flags, final checkpoints (including optimiser states), tokenizer, manifests, source code, benchmark and verification logs, and final evaluations. The raw token arrays and local Parquet shard are not included. The reported results are an analysis of the supplied experiment, not an independent rerun. No complete intermediate validation learning curve is present in this archive.

4. Language-model performance

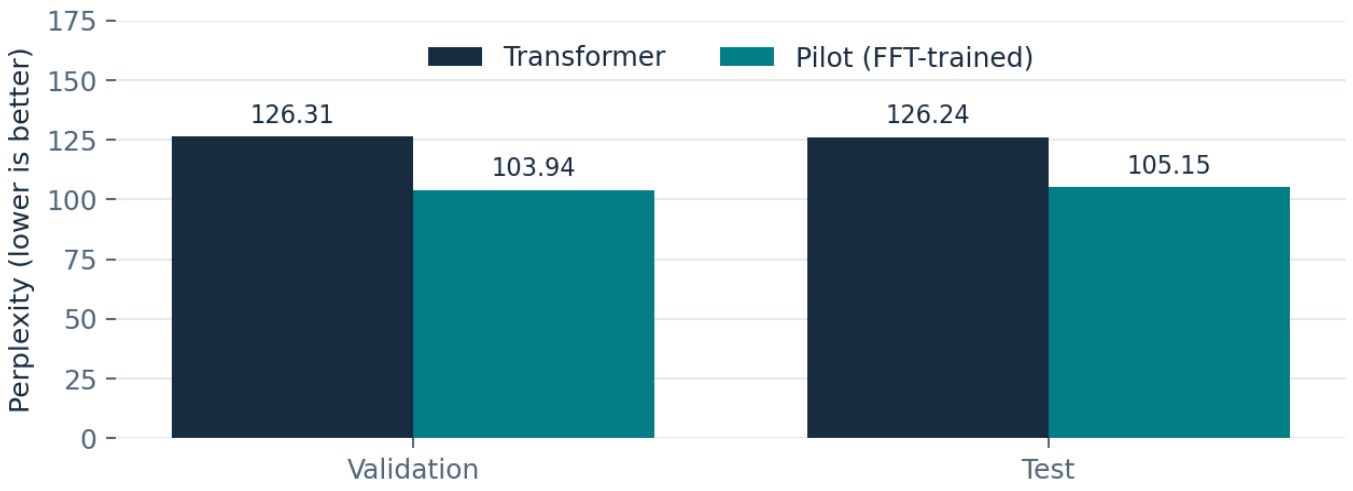


Figure 1. Final held-out perplexity, 8,192 scored targets per split and model. Lower is better.

Metric	Transformer	Pilot FFT
Mean training NLL	5.1739	4.9918
Validation NLL	4.8388	4.6438
Validation perplexity	126.31	103.94
Validation top-1	16.92%	20.32%
Test NLL	4.8382	4.6554
Test perplexity	126.24	105.15
Test top-1	17.61%	20.35%

Pilot lowers test NLL by 0.18275 nats/token and test perplexity by 16.7%. Top-1 accuracy improves by 2.73 percentage points. Validation perplexity is 17.7% lower. The two recorded test windows (4,096 positions each) both favour Pilot: 4.5920 vs 4.7921 and 4.7188 vs 4.8843 nats/token.

Interpretation. This is a positive language-modelling result at equal training exposure and nearly equal parameters. It is not a comparison at equal FLOPs, wall time or energy. Only two aggregate windows per split are available; no reliable across-seed uncertainty estimate or broad generalisation claim follows.

5. Factual recall and reasoning

The frozen checkpoints answer synthetic text prompts using four single-token candidate answers (nominal chance 25%) and unrestricted vocabulary argmax. No probe training, special memory input or additional weight updates are used. The final main suite contains 96 scored rows per model; all are marked OK.

Task	Rows	Transformer	Pilot FFT
Recall, gap 0	8	0/8	0/8
Recall, gap 128	8	2/8	1/8
Binding, 4 facts	8	0/8	0/8
Binding, 8 facts	8	2/8	2/8
Overwrite (both roles)	16	5/16	3/16
Counterfactual (all roles)	32	7/32	3/32
Two-hop retrieval	8	0/8	1/8
Three-hop retrieval	8	1/8	1/8
Total	96	17/96 (17.7%)	11/96 (11.5%)
Unrestricted correct	96	0/96	0/96

Overwrite includes updated and unchanged facts; counterfactual cases include target and unrelated facts before and after an edit. Rows within these groups are related, so the total is descriptive rather than an independent-sample significance test. Both models perform poorly even on immediate recall and four-fact binding. Pilot does not improve on the Transformer across this suite.

What failure does and does not mean

The probes require both interpreting the prompt and retrieving the appropriate association. Failure at zero delay prevents attributing all long-delay errors to forgetting. The small vocabulary, synthetic questions, narrow model width and modest number of examples limit inference. These results do not establish general reasoning ability for either model.

The long-fact audit on the next page separately tests whether additional context changes the same model's answer.

6. Context retention and copying

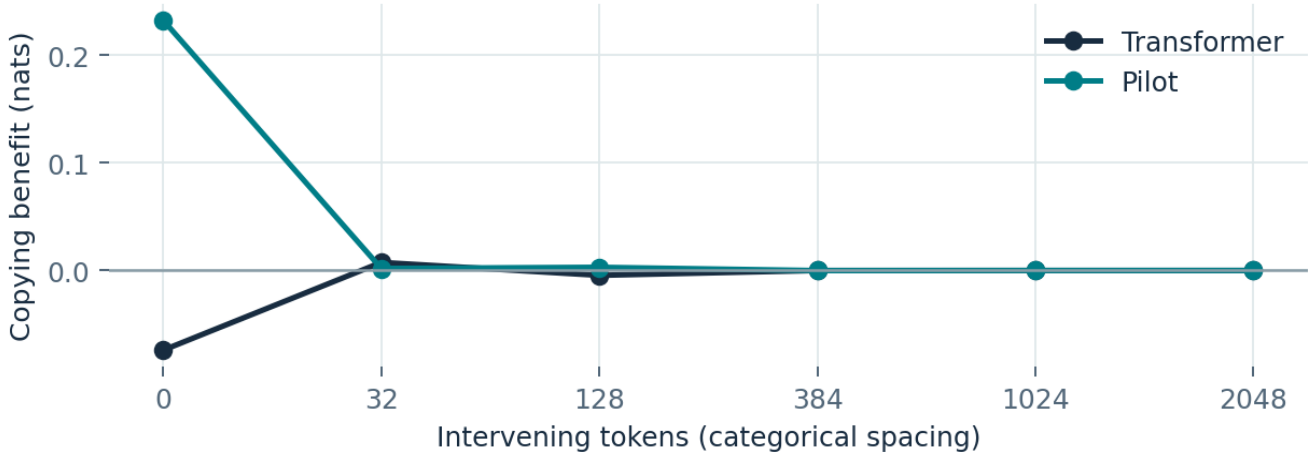


Figure 2. Real-text repeated-passage benefit (control NLL minus repeated NLL), two examples per gap. Higher is better; Transformer inputs are cropped to 256 tokens.

Gap (tokens)	Transformer benefit	Pilot benefit
0	−0.0741	+0.2326
32	+0.00752	+0.00191
128	−0.00450	+0.00298
384	0	−0.000075
1,024	0	+0.00000006
2,048	0	+0.00000075

Pilot's immediate copying benefit fades almost completely by a 32-token gap. At the longest gaps the Transformer's zero benefit is expected: the earlier passage is outside its 256-token input window. Pilot receives the full sequence but shows essentially no benefit either. Two examples per condition make this an exploratory diagnostic, not a precise retention curve.

Long-fact and old-prefix interventions

Across 12 long-fact examples (two at each of 0, 128, 384, 512, 1,024 and 2,048 distractor tokens), Pilot scores 5/12 with full context and 5/12 with the recent 256-token suffix; all paired predictions are identical. The Transformer scores 3/12 with the suffix; its full-context condition is available only for four shorter prompts. All unrestricted answers fail.

The window audit uses two fixed targets. Pilot's mean NLL is 7.0166 at context 256 and 7.0165 at 2,048; no material predictive benefit appears. Replacing old prefixes does change logits on these examples, so the correct conclusion is weak measured benefit, not exact independence from all distant inputs.

7. Efficiency and numerical verification

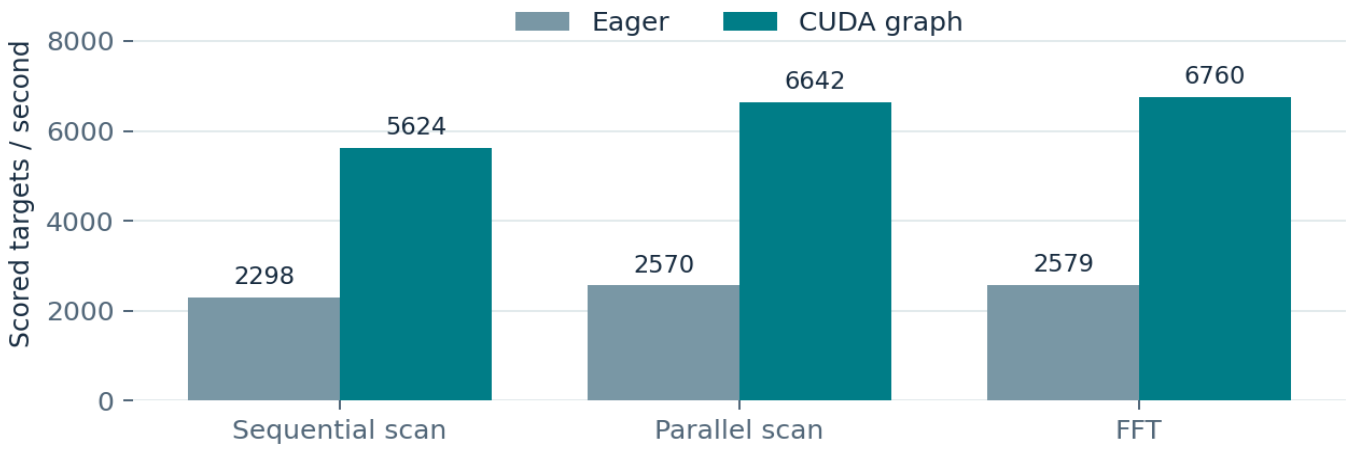


Figure 3. Short full-training-step calibration benchmarks on Tesla T4, 12 timed updates per mode, same batch and model. Pilot FFT is selected.

Full training	Transformer	Pilot FFT
Measured targets/s	405,610	6,776
Training-only time	156.7 s	9,381.0 s
Selected execution	CUDA graph	FFT + CUDA graph

FFT + graph is 2.62× faster than eager FFT in calibration; parallel scan + graph is within 1.7% of FFT + graph. The full Pilot run is still approximately 59.9× slower than the Transformer. Overall recorded elapsed time is 9,815 s (2h 43m 35s), within the four-hour session budget. Timings are implementation-specific, not a theoretical architecture speed limit.

Numerical checks: same mathematics

Recorded check	Maximum absolute error
Float64 history vs direct summation	8.16×10^{-15} (forward); 9.04×10^{-12} (gradient)
FFT graph vs eager FFT, 2 updates	0 (logits, clipped gradients, loss, parameters)
Alternative backends, 2 updates	5.50×10^{-5} logits; 2.59×10^{-5} parameters
Trained FFT vs sequential / parallel	1.24×10^{-5} / 1.34×10^{-5} logits

All recorded comparisons pass their configured tolerances. CUDA graphs capture forward/backward and replay the same operations; AdamW remains eager. Final-checkpoint eager inference at length 128 takes 0.195 s FFT, 0.210 s scan and 0.192 s parallel (three repeats). These checks establish local numerical equivalence, not bitwise-identical long training trajectories.

8. Conclusions and proposed follow-up

What works. Pilot is trainable at this scale, completes the matched exposure and yields better held-out language prediction than the standard Transformer. The FFT, sequential and parallel history calculations are interchangeable to numerical tolerance, and graph execution accelerates the original cell without replacing its equations. These are useful feasibility and quality findings.

What remains open. Neither model solves the evaluation-only recall, binding, overwrite or multi-hop tasks reliably. Pilot shows little useful long-range copying and incurs a large sequential training cost. The fixed decay of 0.9 gives the history a relative-weight half-life of about 6.58 tokens. The recorded mean write gates are 0.599, 0.613 and 0.511 across layers, measured on one 256-token test sequence. These are plausible retention bottlenecks, not proof of the full recurrent Jacobian or a causal explanation of forgetting.

Next experiments

1. Replicate the language result. Use at least three independent seeds, evaluate the full held-out splits or larger independent test samples, and repeat on broader corpora and model sizes. Include a stronger modern positional baseline as a separate control, while retaining the standard Transformer for continuity.

2. Isolate retention. Compare fixed $\lambda = 0.9$ against uniform learned decay and multi-timescale initialisation using the existing eight heads. Keep gates, FFN, data and token budget unchanged. Extend training contexts and pre-register copying, old-prefix and counterfactual tests with many more examples.

3. Measure efficiency fairly. Profile recurrent kernels, explore exact fused execution, report memory and compute, and compare equal wall-clock as well as equal-token budgets. Verify gradients and updates after optimisation changes. Test longer contexts without assuming that longer sequences imply learned memory.

Sources and audit trail

Primary: supplied exilrm_t4x2_head_to_head_240min_20261109_1794336900_results(1).zip. Evidence: plan.json, summary.json, both progress.json and evaluation.json files, probe_results.json, metrics/window_audit.csv, real_repeat.csv, long_fact_audit.csv, backend_audit.csv, history_reference.json, verification_*.json, bench_*.json, data/manifest.json, source manifests, research500.py and exilrm_time_budget.py. Aggregate counts and relative improvements calculated from these files.

Format reference: PHSMLP_Research_Report(1).pdf (10 November 2026), pp. 1–8. Dataset: FineWeb-Edu, sample-10BT (HuggingFaceFW/fineweb-edu). Project lineage: ILRM / Matrix EXILRM; renamed Pilot for this report. No claim of novel FFT convolution, exhaustive prior-art clearance or independent replication.

Bottom line: largely positive as an exploratory architecture test, but not conclusive. Pilot's prediction quality warrants further research; effective long-term memory and practical efficiency remain unsolved.

Appendix A. Pilot block diagram

One layer, one head shown; eight heads operate in parallel.

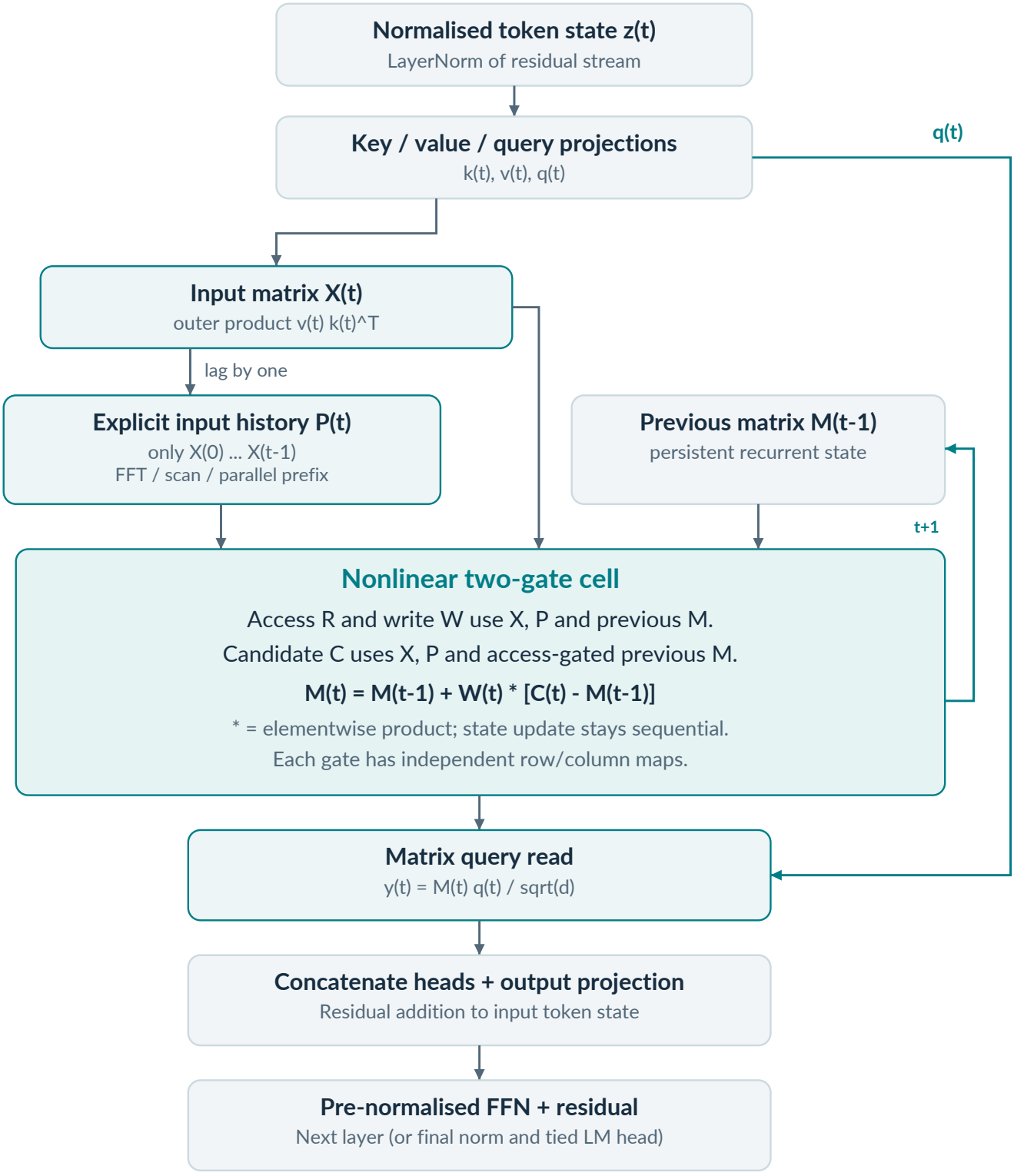


Figure 4. The history is computed from past input matrices, not past hidden states.
The direct input and history feed both gates and candidate; the query bypasses the update.
The feedback arrow carries the updated matrix to the next timestep, not to the history.